WE ARE

DIM3NSION

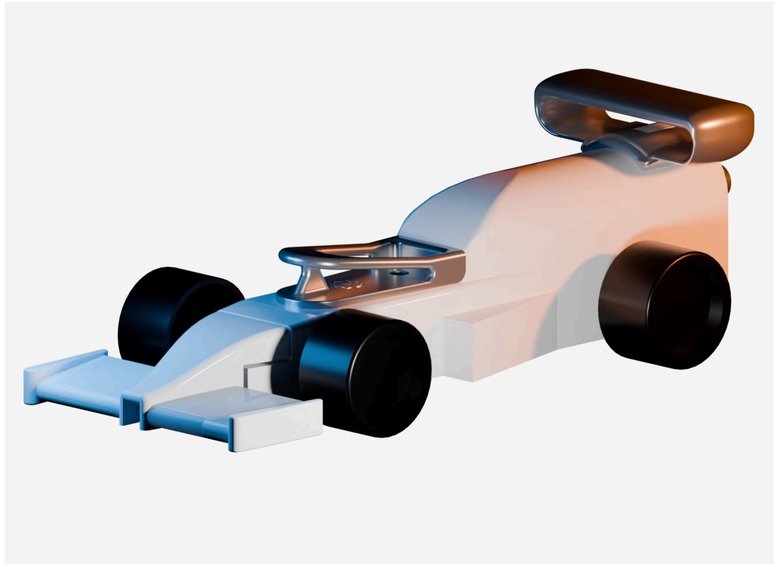
DESIGN AND ENGINEERING PORTFOLIO

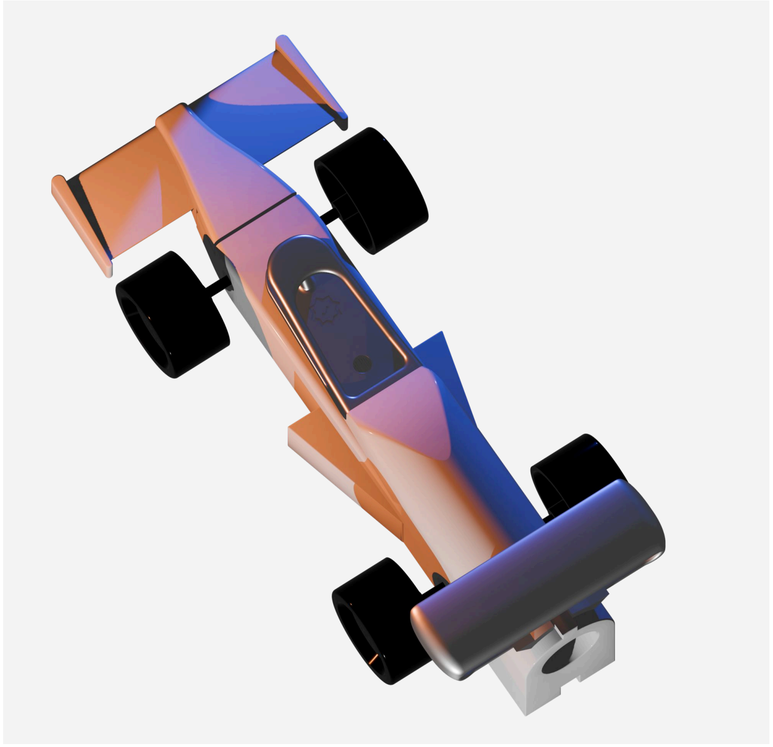
OVERVIEW

Hyp3rsoniX 3.0, our flagship car model, was designed to break aerodybamic and speed records. It incorporated innovative features that make air flow polished and efficient. It's centered on several optimizations: the modeling process was constructed around aero devices that improved drag reduction, air fluidity, and track speed.

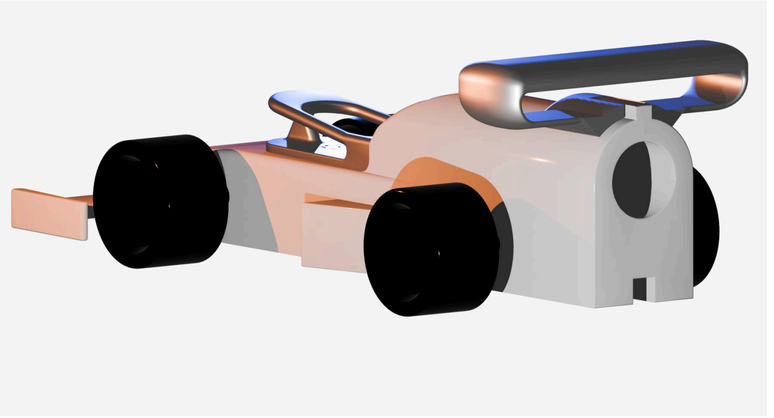
DESIGN RESEARCH

Our team of engineers thoroughly researched key aspects that lead to a high-performance vehicle: drag reduction, pressure points, and velocity, that improve overall air speed over the body of our vehicle; while also focusing on fluid manipulation on surfaces, and turbulence handling, refining air flow shape through the car's chassis. Research was also paired with simulations that rectified the design choices and the aerodynamic principles that our team has followed.





REMARKABLE EFFICIENCY AND OPTIMIZATION AROUND AIRFLOW

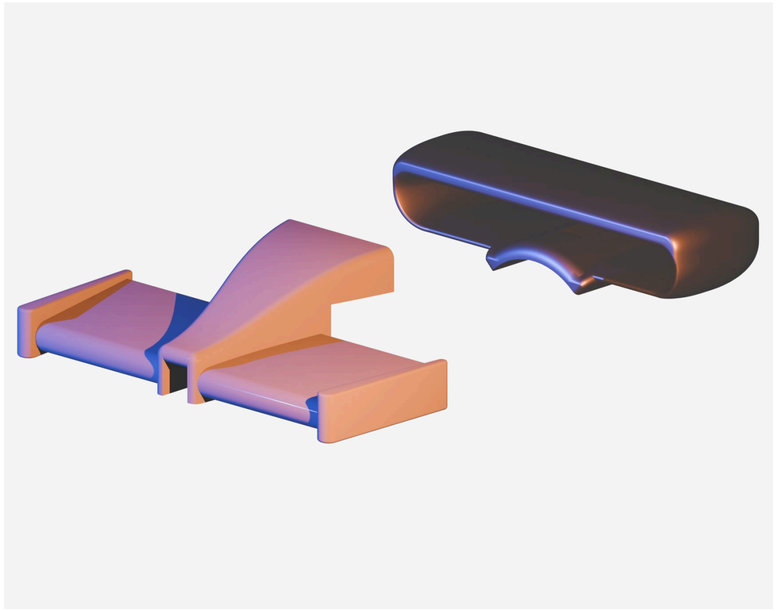


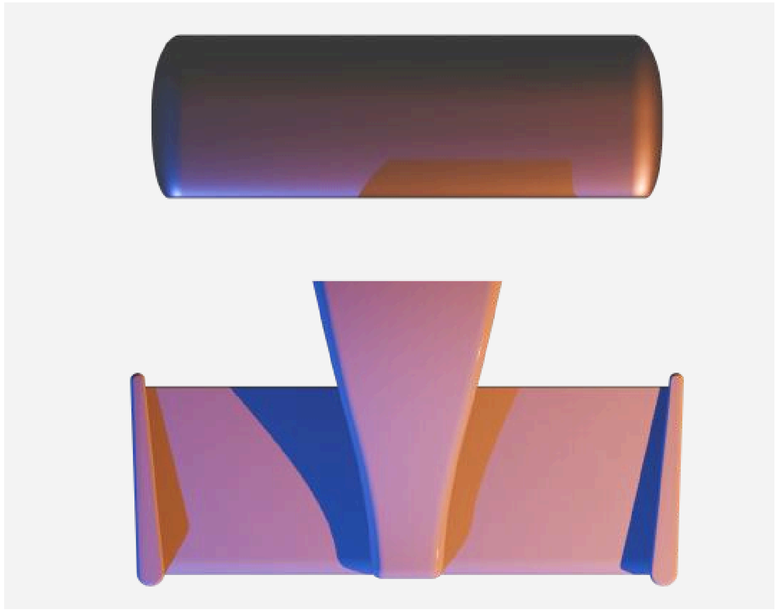
KEY FEATURES

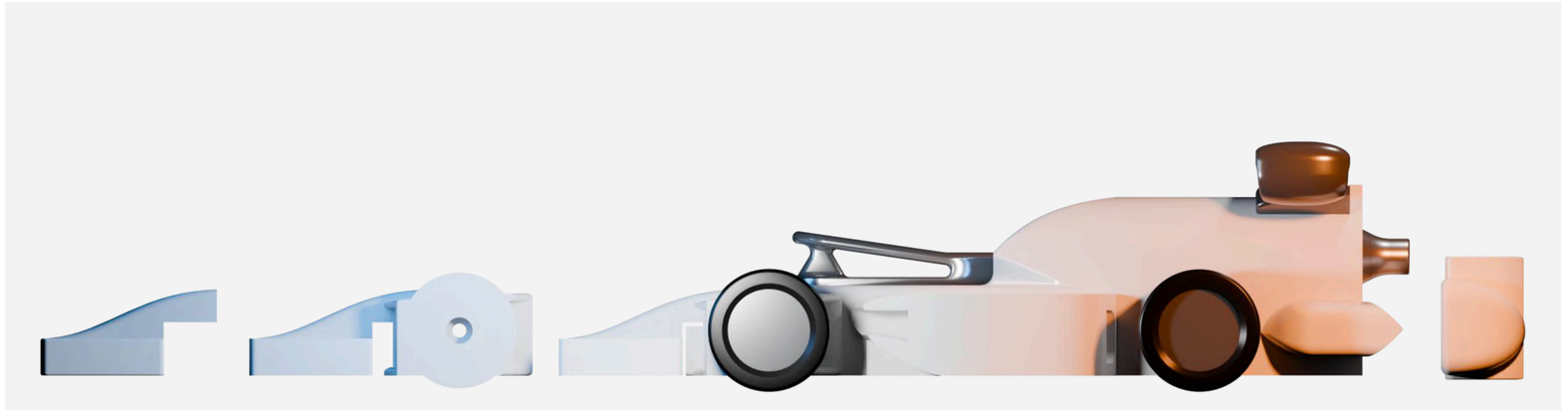
Front Wing: The front wing design facilitates air flow redirection above the wheel, surrounding the fluid around the silhouette of the tires. The end-plates stabilize turbulent air caused by drastic fluid manipulation; it also deflects the air sideways slightly, reinforcing the main concept of diverting fast air around the wheel.

Body: The nose cone and body display a slick, round-edge design; acting as a spearhead piercing fast fluid coming from the front of the car and facilitating the smooth passage of air throughout our car's contour. Adjacent to the halo, the body design resembles the shape of a bullet train, creating a seamless transition between the cockpit and the propulsor chamber.

Rear Wing: The oval-shaped closed-loop rear wing was chosen instead of having open horizontal wingtips with vertical endplates. The top main element curves down into the body, forming a single continuous loop that eliminates traditional exposed wingtips reduces the formation of wingtip vortices, which is a primary cause of induced drag.







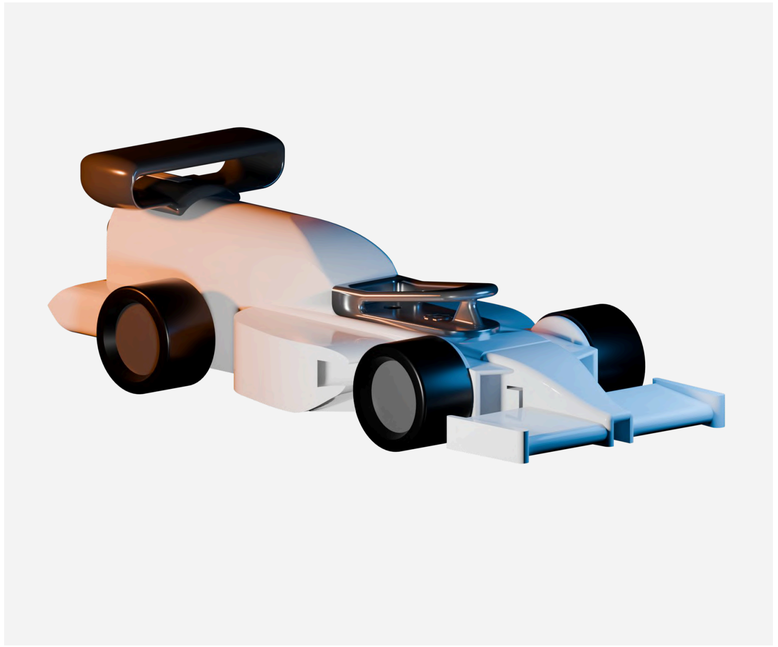
CAD DESIGN PRINCIPLES

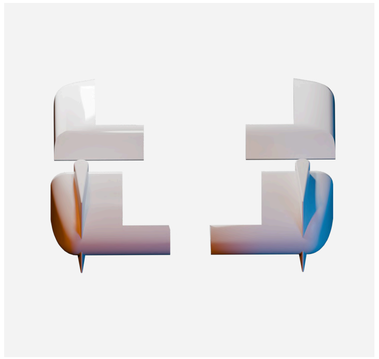
Reducing overall turbulence: To address this challenge, we evaluated a broad range of sidepod and rear diffuser configurations, assessing their impact on airflow stability and wake behavior. While more complex geometries can offer localized performance gains, they also introduce additional surfaces and that may make the flow more difficult to control. Prioritizing design simplicity and maintaining alignment with current Formula 1 development approaches therefore suggests that minimal sidepod and rear diffuser designs are the most effective solution. Their cleaner surfaces help guide airflow more consistently, reduce flow separation, and limit the formation of turbulent air behind the car. This approach supports a more stable aerodynamic platform while also simplifying development, validation, and future design refinement.

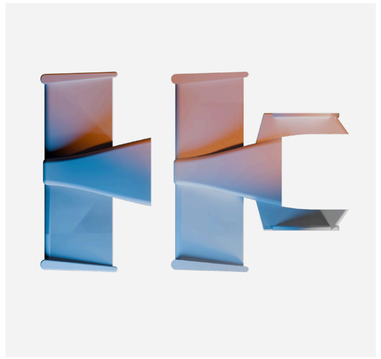
Reducing Total Force (drag): We have incorporated carefully shaped curved surfaces throughout the design to reduce overall aerodynamic drag and improve airflow efficiency. Key examples include the transition between the halo and propulsor, the nose cone, and the beveled edges along the main body. Each feature is aligned with the vehicle's primary contours to encourage smoother, more favorable laminar airflow, reduce abrupt changes in pressure, and minimize flow separation. Together, these refinements help lower resistance and support more efficient vehicle performance.

Incrementing Max Velocity: Increasing maximum velocity is the combined result of the two factors described above. By reducing both aerodynamic drag and turbulence, our vehicle can achieve a higher maximum air velocity. This improvement enhances aerodynamic efficiency, supports stronger acceleration through high-speed sections, and contributes to improved overall on-track performance.

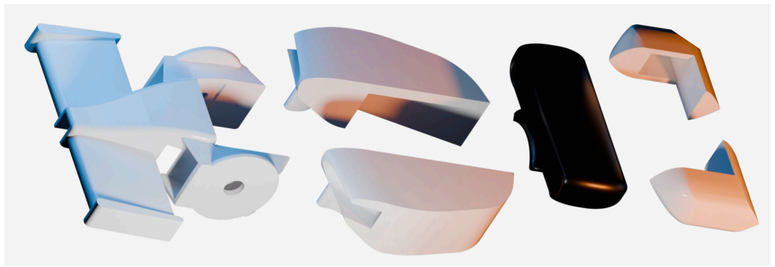
//DESIGN ITERATIONS

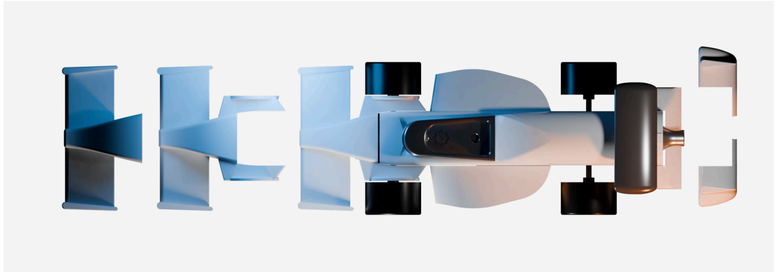






OUR FIRST DESIGNS WERE BASED ON MAXIMUM AIR EFFICIENCY





CAD

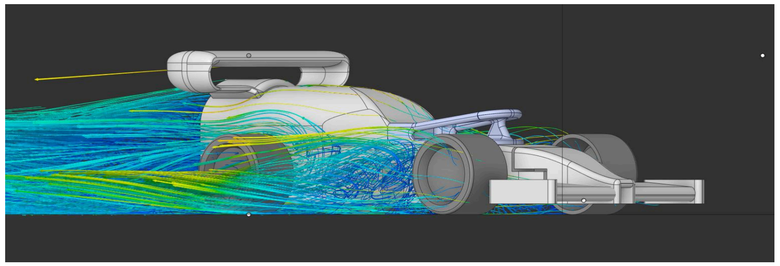
Initially, the car was planned to have complex geometry sidepods, air chambers in the front wing and a diffuser. The sidepod was designed so that air enters into a chamber that stabilized air flow and redirected it inside the small gap between the wheel and the body. The front wing also had chambers that directed the air to the sidepods' inlet, this air passage also served as a turbulence reducer around the wheel, for they are hollow and air can get trapped inside the tires. The diffuser was used to stabilize turbulence in the rear flat part of the vehicle, it also had vertical wings that further reinforced this main idea.

Our current version is significantly simpler and more efficient, reducing fabrication complexity while maintaining strong performance. This was achieved by simplifying the side pods, removing the front-wing chambers, and eliminating the diffuser concept. The lighter design also increased Air Max Velocity from 28 m/s to 29 m/s. In addition, a closer review of the rules revealed that some of our more complex concepts were based on an incorrect interpretation, leading us to remove them from the design.

OUR NEWER DESIGNS ARE LIGHTER AND BETTER OPTIMIZED

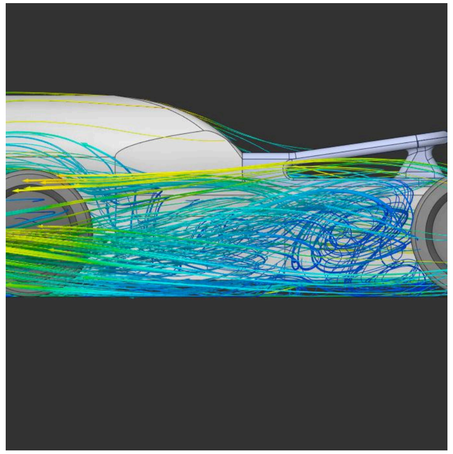
OVERVIEW

The current model outperforms the previous iterations by 1 m/s max velocity (29 m/s current, 28 m/s previous), even though total force is greater (0.7 N current, 0.68 previous). Previous model also had less turbulent zones; however, current model is less efficient at handling turbulence. In the previous design, the side pods were equipped with aero devices to handle the dirty air from the front wheels; conversely, the current design does not incorporate said features, trading off extra weight for more turbulent airflow.



SIDE PODS

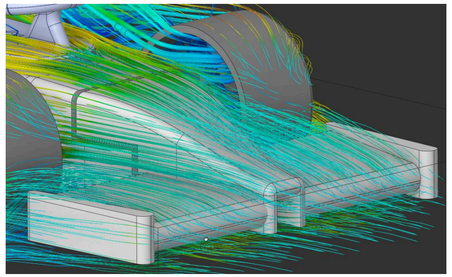
The side pods remained largely unchanged, with only minor modifications. Their primary function is to direct airflow from the front wing along the sides of the vehicle while avoiding additional surfaces that could generate unnecessary aerodynamic drag. This approach prioritized reduced total force and mass, along with increased maximum velocity, over minimizing turbulence.



FRONT WING

The front wing demonstrates effective air deflection by shaping the airflow to follow the wheel's silhouette. Its endplates help prevent high-pressure air from spilling downward into the low-pressure region, maintaining a clear separation between the two pressure zones. At the same time, they redirect the airflow laterally, guiding it away from the wheel to reduce unwanted interaction with the rotating surface.

As demonstrated in the simulation below, the airflow reaches relatively high speeds. The front wing directs this airflow smoothly through the wheel area, as illustrated by the yellow vector lines.



REAR WING

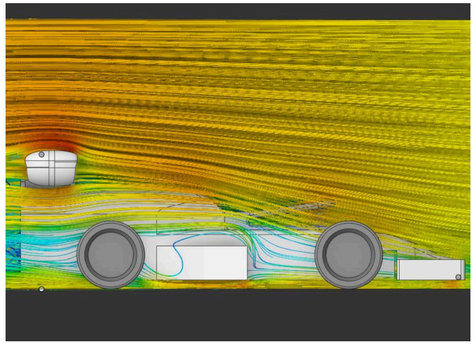
The closed-loop configuration is the most effective design choice because it minimizes the likelihood of generating wing-tip vortices, which are a major source of turbulent airflow. In addition, its oval geometry enables air to move smoothly and continuously with minimal obstruction. This streamlined flow helps maintain consistent speed while improving the overall aerodynamic efficiency of the design.

As shown in the image, the airflow behind the vehicle is traveling at very high speed. This indicates that the rear wing generates minimal drag while still producing sufficient downforce to maintain tire grip, stability, and control.

AERODYNAMIC ANALYSIS LEADS DESIGN DECISIONS

BODY

The body design is based on a streamlined, bullet-like chassis. Its beveled surfaces help redirect incoming air around the body, maintaining smooth and consistent airflow across the exterior. The sleek, curved form efficiently channels air from the front toward the sides, reducing disruption and supporting strong aerodynamic performance.



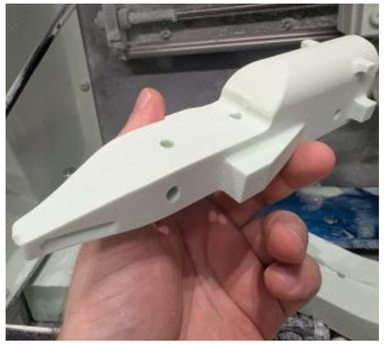
A key advantage of this geometry is its ability to reduce the total frontal cross-sectional area while maintaining structural integrity. By minimizing the cross-sectional profile, the design directly reduces overall drag force, which can improve aerodynamic efficiency and performance.

Manufacturing curved geometries significantly improves CNC machining efficiency by enabling continuous, fluid tool paths that reduce spindle chatter and tool wear compared to sharp, abrupt transitions. Additionally, these smooth surfaces simplify post-processing—distributing sanding pressure evenly and preventing liquid sealants or spray paint from pulling away from sharp edges, which ensures a uniform finish and stronger structural durability.

//MANUFACTURING

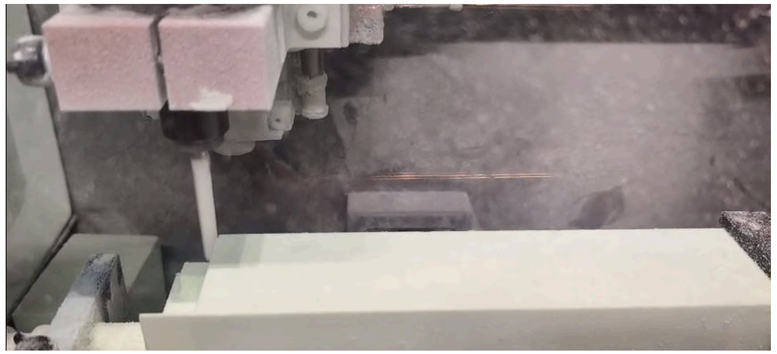
DIM3NSION

CAD



Fabrication of the main chassis began with the submission of our 3D model files for machining on a high-precision CNC milling machine. The raw material stock was supplied directly by the STEM Racing Panamá organization, ensuring that the chassis met the official competition standards and material requirements throughout the manufacturing process.

Utilizing CNC machining allowed for precise translation of our complex aerodynamic curves from the digital CAD environment into the physical polyurethane/balsa block.



SURFACE FINISHING

To eliminate surface rough spots left by the milling bit and minimize aerodynamic skin friction drag, the chassis underwent a hand-sanding and machine-sanding process. Grit sandpapers were used to smooth out tool marks, and a rotary tool to perform heavy polishing. This was made to create a low-friction base layer prior to sealing and painting.

3D PRINTING

The front and rear wings were manufactured from PLA (polylactic acid) using 3D printing. This manufacturing process enabled the production of precise, customized geometries and complex aerodynamic profiles that would have been difficult to achieve through milling.

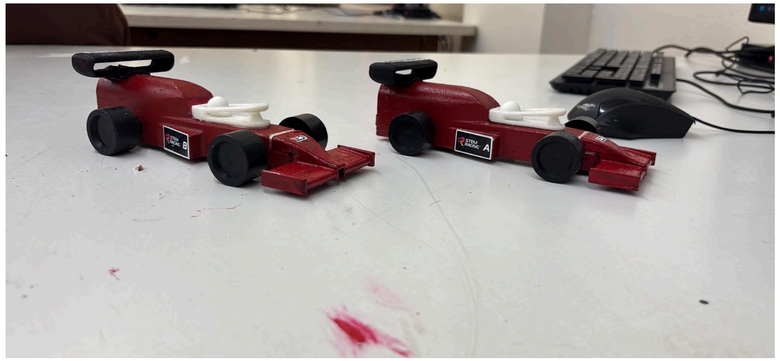
A layer height of approximately 0.28 mm was selected to provide a suitable balance between surface detail, print efficiency, and dimensional accuracy. The infill density was optimized at approximately 20%, providing sufficient structural rigidity while minimizing overall weight and material consumption.

CHASSIS SEALING

Because porous foam chassis materials have a sand-like finish, substances like glue and paint would be poorly applied; a specialized liquid elastomeric waterproofing layer was applied. This coat effectively seals the surface and adds a glossy finish, forming a barrier against moisture and creating an ultra-smooth, uniform foundation for the final color coats.

PAINT & ASSEMBLY

Once sealed, all custom components, including the main body and aerodynamic wings, were prepped for spray painting. Multiple thin layers of paint were applied to guarantee an even aesthetic finish without adding excess weight. Assembly was performed using high-strength wood glue to secure the 3D-printed wings seamlessly to the main chassis body without requiring heavy mechanical fasteners.



LIVERY AND TOLERANCES

The final aesthetic stage involved placing team livery stickers and sponsor decals precisely onto designated body zones. Throughout the entire manufacturing and assembly sequence, strict dimensional checks were performed using digital calipers. Critical dimensions and component fits were consistently held to a tight tolerance of approximately 1.0 mm, ensuring full adherence to competition technical regulations and optimal aerodynamic alignment.

FINAL MODEL

By integrating CAD design with high-precision CNC machining, the chassis transitions efficiently from an optimized aerodynamic model into a fully realized physical component. This integrated workflow enables the milling process to reproduce every shape, and surface bevel with exceptional accuracy.

As a result, the chassis achieves the structural integrity required to comply with strict competition regulations while preserving the precise dimensional tolerances essential to peak aerodynamic performance.

DRIVE